A field guide to analyzing neural and behavioral time-series data

Always carry a prediction.

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About this course

Comprehensive guide to analyzing time-series data in neuroscience

For anyone interested in working with modern neuroscience data

<u>Whether you want to</u>

Analyze data generated by a specific experiment addressing a specific question

Extract new scientific results from existing data

<u>Course components</u>

Time-series fundamentals

Philosophy of analyzing scientific data soundly and efficiently

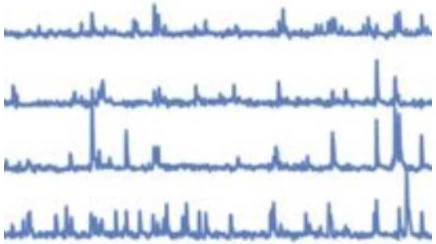
Practical advice for analyzing real datasets

Methods survey

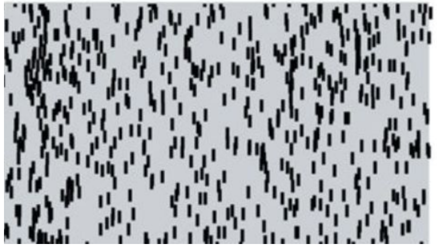
<u>Designed to be used as both course and guidebook/reference</u>

Why is analyzing time-series data important?

Default output of most modern measurement devices



Adapted from de Groot et al, *eLife* 2020 (CC-BY 4.0)



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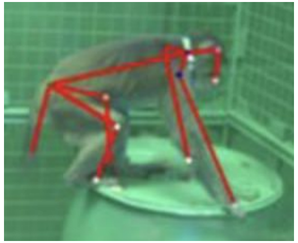

Adapted from Tang et al, *eLife* 2025 (CC-BY 4.0)

Crucial but not easy to understand how to work rigorously with such data.

Analysis is much more than classical signal processing.

Requires skillful integration of domain and technical knowledge.

Why is analyzing biological time-series data hard?

Weird statistics/ lack of trials

Missing data/variable trial lengths

High dimensionality

Violates many assumptions of classic signal processing

Model-fitting can be highly complex

Difficult to interpret analysis results

Many methods to choose from

Big datasets

Multi-modal datasets

Extensive small decisions to make along the way

Easy to make mistakes

Takes time to be rigorous

Goals of this course

Overall:

Fast-track your way to being able to make original,
data-driven contributions to neuroscience research

Along the way:

Understand fundamentals of time-series data

Gain familiarity with canonical and state-of-the-art methods

Gain practical tips & tricks and solutions to common issues

Learn to systematically transform data into results

Learn to design new analyses to ask new questions

The complete outline: everything you need to hit the ground running

II. Fundamental concepts
- Data types in neuroscience
- Mathematical models
- Discriminative vs generative models
- Parameters, fitting and overfitting
- Model comparison
- Random processes perspective
- Dynamical systems perspective

III. Philosophy
- Fundamental theorem of data analysis
- Frameworks, theories, and models
- Descriptive, mechanistic, normative theories
- Theory- vs data-driven approaches
- The value of predictions
- Transforming data into science
- McNamara fallacy

IV. Common issues
- Lack of trials
- Controlled vs naturalistic data
- Nonstationarity/long timescales
- Trials/sessions/animals/conditions
- Binning
- Missing data
- Too much data
- Multicollinearity
- Gotchas: normalization, correlated training/test data, what is N?
- Interpreting data-driven analysis results

V. Practical tips and tricks
- Approaching/vetting a new dataset
- Data pre-processing
- Data munging/storage
- Control datasets
- Desconstructing fit models
- Statistical testing
- How to not make mistakes
- Leveraging LLMs
- Designing custom analyses
- Coding strategies
- Reproducibility
- Data sharing

VIa. Canonical Methods – Pre-processing
- Filtering/smoothing
- Detrending
- Spike sorting
- Estimating firing rates
- Image processing/ROI extraction
- Behavior keypoint tracking
- Segmenting

VIb. Canonical Methods – Classical Signal processing
- Correlation functions
- Fourier transforms and power spectra
- Linear filters and impulse response

VIc. Canonical Methods – Neural Data Analysis
- Raster plots and inter-spike interval distributions
- Peristimulus time histograms (PSTHs)
- Tuning curves
- Linear-nonlinear-Poisson (LNP) and generalized linear models (GLM)
- Spike-triggered average (STA)

VII. Advanced methods
- Dimensionality reduction
- Clustering/segmentation
- Statistical models
- Latent variable models
- Dynamics/state-space models
- Mechanistic models
- Connectome-based models
- Inferring connectivity
- Spatiotemporal analyses
- Artificial/deep neural networks
- Variational inference/ELBO
- Extended learning/inference techniques
- Information theory
- Validation metrics

VIII. Miscellaneous
- Other methods
- Mathematical/coding background recommendations
- Common software packages
- Other courses/resources
- Frequently asked questions